

[445] A MEMOIR ON QUARTIC SURFACES. (CONTINUED)

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(Continuation from book p. 153; articles 56–112.)

56. We may through 3 draw the lines LM , QT to meet 14, 26 and 12, 46 respectively; and through 5 the lines RS , NP to meet 14, 26 and 12, 46 respectively. Observe that the points O in the figure are apparent intersections only; viz., NP does not meet QT , nor LM meet RS . In fact, if NP met QT it would be a line in the series of lines meeting 14, QT , 26; or 5 would be situate in a hyperboloid, determined by means of the points 1, 2, 4, 6, 3; viz., 5 would not be an arbitrary point: and so LM does not meet RS . Now the quadrics E , H meet in the lines 14, 26, LM , NP , and the quadrics A , C in the lines 12, 46, QT , RS . Suppose that we had identically $(* \$ A, C, E, H)^2 = 0$; putting therein $E = 0$, $H = 0$, we should have $(* \$ A, C)^2 = 0$, viz.,

$$(A + \lambda C)(A + \mu C) = 0;$$

or there would exist quadrics of the forms $A + \lambda C = 0$ containing the lines 14, 26, LM , NP . Now there is no quadric surface $A + \lambda C = 0$ containing the line NP ; for $A + \lambda C = 0$ is a quadric containing the sides of the quadrilateral $QRST$; the generating lines of the one kind meet each of the lines RS , QT ; those of the other kind neither. Hence NP , which meets RS but not QT , cannot be a generating line of either kind; and we have no identical relation $(A, C, E, H)^2 = 0$.

57. In the octadic surface with 9 nodes: starting with any 7 nodes of the octad, 9 is not the 8th point of the octad, and hence (by the theory of the dianome) it must lie in the dianodal surface of the 7 points; that is, the dianodal surface of the 7 points must pass through 9, viz., through any point whatever of the Jacobian curve of the 7 points, that is, of the octad; or (what is the same thing) the dianodal surface of the 7 points passes through the Jacobian curve of the octad. This is an obvious property of the dianodal surface, the surface $J(P, Q, R, V) = 0$ contains the Jacobian curve $J(P, Q, R) = 0$. But it further appears that, starting with any 6 points of the octad and with the point 9 (that is, any point whatever of the Jacobian curve), the dianodal surface of these 7 points must contain the remaining 2 points of the octad. And in the octadic surface with 10 nodes, starting with any 5 points of the octad and with the points 9 and 10 (that is, any two points on the Jacobian curve) the dianodal surface of these 7 points must contain the remaining three points of the octad. I have not attempted to verify these last properties of the dianodal surface.

Dianomes with 9 or 10 Nodes.

58. I now consider the dianomes with 9 and 10 nodes. Starting from the general form

$$(a, b, c \times P, Q)^2 + \theta \Delta = 0,$$

where $\Delta = 0$ is a particular quartic surface having the 8 nodes, it at once appears that if there is a 9th node, say 9, the node must be a point on the Jacobian curve $J(P, Q, \Delta) = 0$; hence, taking it to be one such point, the number of constants in the resulting equation should be 1. Hence if $P = 0$ be the quadric surface through the 9 points, and $\Delta = 0$ a particular quartic surface having the 9 points as nodes, the general equation is $\alpha P^2 + \theta \Delta = 0$.

59. But we may consider the question somewhat differently. Starting with the 7 given points 1, 2, 3, 4, 5, 6, 7 and with 8 a given point on the dianodal surface of the 7 points; it is clear that 9 must be on the dianodal surface 1234567 and also on the dianodal surface 1234568; the complete intersection is of the order 36, and we have to consider how this breaks up so as to contain a part of itself the dianodal curve of the order 18.

Dianodal Curve of 8 Points.

60. Consider first any 8 points whatever 1, 2, 3, 4, 5, 6, 7, 8 where 8 is not on the dianodal surface 1234567, nor 7 on the dianodal surface 1234568. The two surfaces have in common the 15 lines 12, 13, ..., 56 and the skew cubic 123456, they therefore besides intersect in a curve of the order 18. At the point 1 the tangent cubic cones of the two surfaces intersect in the lines 12, 13, 14, 15, 16 and the tangent to the skew cubic 123456, 6 lines lying in a quadric cone; they therefore besides intersect in 3 lines lying in a plane; that is, the point 1 is on the curve of the order 18 an actual triple point, the 3 tangents lying *in plano*; and the like of course in regard to each of the points 2, 3, 4, 5, 6. But as 7, 8 lie each of them on only one of the two surfaces, the curve of the order 18 does not pass through 7 or 8.

61. If, however, 8 lies on the dianodal surface 1234567, then each of the 8 points will lie on the dianodal surface of the other 7; and in particular 7 will lie on the dianodal surface 1234568. The surfaces intersect as before in a residual curve of the order 18; the only difference is that 7 and 8 are now points on each surface; viz., each of them is on one of the surfaces an ordinary point, and on the other a cubic node; the points 7 and 8 are thus each of them an actual triple point on the curve; and at each of them the 3 tangents are *in plano*. We thus see that the dianodal curve 12345678 is a curve of the order 18, such that each of the 8 points is a triple point on the curve, the tangents at each of them being *in plano*.

Ten Nodes.

62. Suppose there is a 10th node, say 10; starting from the equation $\alpha P^2 + \theta \Delta = 0$ (P = the quadric surface through the 9 points, Δ = a particular quartic surface having the 9 points as nodes), it at once appears that the node must be one of the points $J(P, \Delta) = 0$; hence, taking it to be one of these points, we have 4 equations, which, in virtue of the node being one of the points in question, reduce themselves to a single equation determining the ratio $\alpha : \theta$; we have thus a completely determinate surface, say $\square = 0$ having the 10 points as nodes. The number of points $J(P, \Delta)$, writing in the formula No. 5, $a = 1$, $b = 3$, is obtained as $1 + 3 + 9 + 27 = 40$, but it is to be observed that the surface $P = 0$ passes through each of the 9 nodes of the surface $\Delta = 0$; these count twice among the points $J(P, \Delta) = 0$, and the number of residual points (or say the dianodal centres of the 9 points) is $40 - 18 = 22$; viz., this is the number of positions of the node 10. [The nine points count each three times, and the number of residual points, or positions of the node 10, is thus not $40 - 18 = 22$, but $40 - 27 = 13$.]

Dianodal Centres of 9 Points.

63. In further explanation, observe that 9 is any point on the dianodal curve 12345678; the node 10 must lie on this same curve, and also on the dianodal surface 1234569. Take $P = 0$ the quadric through all the 9 points, $Q = 0$ a quadric through all but the point 9, $R = 0$ through all but the point 8, $S = 0$ through all but the point 7. The dianodal curve 12345678 is $J(P, Q, V) = 0$, and the dianodal surface 1234569 is $J(P, R, S, V) = 0$; the total number

of intersections is $6 \times 18 = 108$; these include the $4 \times 18 = 72$ points of intersection of the dianodal curve $J(P, Q, \Delta) = 0$ with the Jacobian surface $J(P, Q, R, S) = 0$, except the four points $J(P, Q) = 0$, which are the vertices of the 4 quadric cones through 1, 2, 3, 4, 5, 6, 7, 8 (which 4 points are not situate on the curve $J(P, R, S) = 0$), and there are besides 40 points $\{108 = (72 - 4) + 40\}$ which are the before-mentioned points $J(P, \Delta) = 0$; viz., these are the 9 points each twice [three times], and the residual 22 [13] points which are the dianodal centres of the 9 points.

General result as to the Dianomes.

64. We have thus established the theory of the dianome quartic surfaces; viz., we have

- The octodianome, 8 nodes, 7 of them arbitrary, and the 8th an arbitrary point on the dianodal surface (order 6) of the 7 points.
- The enneadianome, 9 nodes, the 9th an arbitrary point on the dianodal curve (order 18) of the 8 points.
- The decadianome, 10 nodes, the 10th any one of the 22 [13] dianodal centres of the 9 points.

And as already mentioned, so long as the first 7 nodes are arbitrary, there cannot be more than 10 nodes in all.

The Symmetroid.

The Lineolar Correspondence of Quartic Surfaces.

65. I consider four equations $S = 0, T = 0, U = 0, V = 0$, lineolar in regard to the two sets of coordinates (x, y, z, w) and $(\alpha, \beta, \gamma, \delta)$; viz., each of these equations is of the form

$$(*\$x, y, z, w \times \alpha, \beta, \gamma, \delta) = 0.$$

This implies that the point (x, y, z, w) lies on a certain quartic surface $\Theta = 0$, and the point $(\alpha, \beta, \gamma, \delta)$ on a certain quartic surface $\Delta = 0$, and that the two surfaces correspond point to point to each other. In fact, writing the four equations in the form

$$\begin{aligned} L\alpha + M\beta + N\gamma + P\delta &= 0, \\ L'\alpha + M'\beta + N'\gamma + P'\delta &= 0, \\ L''\alpha + M''\beta + N''\gamma + P''\delta &= 0, \\ L'''\alpha + M'''\beta + N'''\gamma + P'''\delta &= 0, \end{aligned}$$

where $L, \&c.$, are linear functions of (x, y, z, w) , then eliminating $(\alpha, \beta, \gamma, \delta)$, we obtain the equation

$$\Theta = \begin{vmatrix} L & M & N & P \\ L' & M' & N' & P' \\ L'' & M'' & N'' & P'' \\ L''' & M''' & N''' & P''' \end{vmatrix} = 0;$$

and similarly, writing the four equations in the form

$$\begin{aligned} Ax + By + Cz + Dw &= 0, \\ A'x + B'y + C'z + D'w &= 0, \\ A''x + B''y + C''z + D''w &= 0, \\ A'''x + B'''y + C'''z + D'''w &= 0, \end{aligned}$$

where $A, \&c.$, are linear functions of $(\alpha, \beta, \gamma, \delta)$, then eliminating (x, y, z, w) , we obtain the equation

$$\Delta = \begin{vmatrix} A & B & C & D \\ A' & B' & C' & D' \\ A'' & B'' & C'' & D'' \\ A''' & B''' & C''' & D''' \end{vmatrix} = 0.$$

Moreover, Θ being $= 0$, the four linear equations in $(\alpha, \beta, \gamma, \delta)$ are equivalent to three equations, and give for instance $(\alpha, \beta, \gamma, \delta)$ proportional to the determinants formed with the matrix

$$\begin{vmatrix} L' & M' & N' & P' \\ L'' & M'' & N'' & P'' \\ L''' & M''' & N''' & P''' \end{vmatrix};$$

and similarly, Δ being $= 0$, the four linear equations in (x, y, z, w) are equivalent to three equations, and give for instance (x, y, z, w) proportional to the determinants formed with the matrix

$$\begin{vmatrix} A' & B' & C' & D' \\ A'' & B'' & C'' & D'' \\ A''' & B''' & C''' & D''' \end{vmatrix},$$

which establishes the point-to-point correspondence of the two surfaces.

66. It would at first sight appear that any quartic surface $(*\$ \alpha, \beta, \gamma, \delta)^4 = 0$ whatever might have its equation expressed in the foregoing determinant form $\Delta = 0$. This equation seems, in fact, to contain homogeneously as many as 64 constants. But if we multiply the determinant line into line by a constant determinant

$$\begin{vmatrix} a & b & c & d \\ a' & b' & c' & d' \\ a'' & b'' & c'' & d'' \\ a''' & b''' & c''' & d''' \end{vmatrix}$$

and then column into column by another constant determinant, the coefficients, all but one of them, of these constant determinants may be used to specialize the form of the resulting equation, [say they are apoclastic constants]; this equation will really contain $64 - (2 \cdot 16 - 1) = 33$ constants; and in order that the quartic surface $(*\$ \alpha, \beta, \gamma, \delta)^4 = 0$ may have its equation expressible in the form $\Delta = 0$, a single relation must hold good among the coefficients; but this in passing¹.

67. Returning to the quartic surface

$$\Delta = \begin{vmatrix} A & B & C & D \\ A' & B' & C' & D' \\ A'' & B'' & C'' & D'' \\ A''' & B''' & C''' & D''' \end{vmatrix} = 0,$$

we may connect this not only with the foregoing surface $\Theta = 0$, but in a similar manner with another quartic surface $\Phi = 0$; viz., taking the current coordinates $(\xi, \eta, \zeta, \omega)$, we may form the

¹Applying the same reasoning to a cubic determinant $\Delta = 0$, the number of constants is $36 - (2 \cdot 9 - 1) = 19$; so that a cubic surface is expressible in the form in question. And so for the quadric determinant $\Delta = 0$, the number of constants is $16 - (2 \cdot 4 - 1) = 9$; so that a quadric surface is expressible in the form in question, as is otherwise obvious.

linealinear equations

$$\begin{aligned} A\xi + A'\eta + A''\zeta + A'''\omega &= 0, \\ B\xi + B'\eta + B''\zeta + B'''\omega &= 0, \\ C\xi + C'\eta + C''\zeta + C'''\omega &= 0, \\ D\xi + D'\eta + D''\zeta + D'''\omega &= 0, \end{aligned}$$

which, by the elimination of $(\xi, \eta, \zeta, \omega)$, give $\Delta = 0$, and by the elimination of $(\alpha, \beta, \gamma, \delta)$ a determinant quartic equation $\Phi = 0$ between the coordinates $(\xi, \eta, \zeta, \omega)$; and of course the two surfaces $\Delta = 0$, $\Phi = 0$ have a point-to-point correspondence such as exists between the surfaces $\Theta = 0$, $\Delta = 0$. The relation of the point $(\alpha, \beta, \gamma, \delta)$ on the surface $\Delta = 0$ to the point (x, y, z, w) on the surface $\Theta = 0$, and to the point $(\xi, \eta, \zeta, \omega)$ on the surface $\Phi = 0$, may be conveniently indicated by means of the diagram

x	y	z	w	
A	B	C	D	ξ
A'	B'	C'	D'	η
A''	B''	C''	D''	ζ
A'''	B'''	C'''	D'''	ω

$\Theta / \Delta / \Phi.$

68. It is to be observed that, writing for $A, B, \dots$ their values as linear functions of $(\alpha, \beta, \gamma, \delta)$, we have in all 64 constant coefficients, which we may conceive arranged in the form of a cube, thus

$$\begin{array}{cccc} a & b & c & d \\ a' & b' & c' & d' \\ \vdots & \vdots & \vdots & \vdots \\ a_1 & b_1 & c_1 & d_1 \\ a'_1 & b'_1 & c'_1 & d'_1 \\ \vdots & \vdots & \vdots & \vdots \end{array}$$

and taking these in fours height-wise, (a, a_1, a_2, a_3) , &c., we compose with them the linear functions $a\alpha + a_1\beta + a_2\gamma + a_3\delta$, &c., which enter into the equation $\Delta = 0$; taking them in fours length-wise, (a, b, c, d) , &c., we compose the linear functions $ax + by + cz + dw$, &c., which enter into the equation $\Theta = 0$; and taking them in fours breadth-wise (a, a', a'', a''') , &c., we compose the linear functions $a\xi + a'\eta + a''\zeta + a'''\omega$, &c., which enter into the equation $\Phi = 0$.

69. The process may be indefinitely repeated; we obtain always the same three surfaces over and over again, but on them an indefinite series of corresponding points; viz., we may write

$$\begin{aligned} \dots \Theta, \Delta, \Phi, \Theta, \Delta, \Phi, \Theta, \Delta, \Phi \dots \\ \dots P_1, Q_1, R_1, P, Q, R, P', Q', R' \dots \end{aligned}$$

viz., a point Q on Δ corresponds to a point P on Θ and to a point R on Φ ; R corresponds to Q on Δ and to a new point P' on Θ ; P' to R on Φ and to a new point Q' on Δ , and so on. And in the opposite direction P corresponds to Q on Δ and to a new point R_1 on Φ ; R_1 to P on Θ and to a new point Q_1 on Δ and so on. And of course the correspondence of any two points of the series, whether belonging to the same surface or to different surfaces, is a one-to-one correspondence.

The Symmetrical Case; Symmetroid and Jacobian.

70. I have established the foregoing general theory; but it is only a particular case of it which connects itself with the theory of nodal quartics; viz., the cube of coefficients is a symmetrically arranged cube

$$\begin{array}{cccc} a & h & g & l \\ h & b & f & m \\ g & f & c & n \\ l & m & n & d \end{array} \quad \begin{array}{ccc} a_1 & h_1 & \dots \\ \vdots & & \end{array}$$

or say its upper face is the symmetrical square matrix

$$\begin{vmatrix} a & h & g & l \\ h & b & f & m \\ g & f & c & n \\ l & m & n & d \end{vmatrix}$$

and the other horizontal planes, the like squares with the several terms affected by suffixes. The surface $\nabla = 0$ is here a surface of the form

$$\nabla = \begin{vmatrix} A & H & G & L \\ H & B & F & M \\ G & F & C & N \\ L & M & N & P \end{vmatrix} = 0$$

$[A, B, \&c. \text{ linear functions of } (\alpha, \beta, \gamma, \delta)]$; viz., ∇ is a symmetrical determinant; I call this a *symmetroid*; the surfaces $\nabla = 0$, $\square = 0$ are one and the same surface, the Jacobian of 4 quadric surfaces; moreover the points P and R are one and the same point, and the correspondence R to P' is a reciprocal one; so that, instead of the indefinite series $\dots \Delta, \Theta, \Delta, \Theta, \Delta \dots$, $\dots P', Q, P, Q, P \dots$, we have a surface, the *symmetroid* $\nabla = 0$ which is a surface with 10 nodes, which is clearly not octadic, and which is therefore the decadianome.

71. Consider the quadric surfaces

$$\begin{aligned} S &= (a, b, c, d, f, g, h, l, m, n \mid x, y, z, w)^2 = 0, \\ T &= (a_1, \dots) \quad \quad \quad , \quad y = 0, \\ U &= (a_2, \dots) \quad \quad \quad , \quad y = 0, \\ V &= (a_3, \dots) \quad \quad \quad , \quad y = 0, \end{aligned}$$

and a point $(\alpha, \beta, \gamma, \delta)$ in the same or in a different space, such that the surface

$$\alpha S + \beta T + \gamma U + \delta V = 0$$

is a cone, or say for shortness,

$(\alpha, \beta, \gamma, \delta)$ is said to be the determining point, or determinant of the cone.

And if we establish the equations

$$\begin{aligned} \partial_x(\alpha S + \beta T + \gamma U + \delta V) &= 0, \\ \partial_y(\quad \text{do.} \quad) &= 0, \\ \partial_z(\quad \text{do.} \quad) &= 0, \\ \partial_w(\quad \text{do.} \quad) &= 0, \end{aligned}$$

which express that the surface is a cone, then (x, y, z, w) is the vertex of the cone. We have thus 4 equations linear in (x, y, z, w) and also in $(\alpha, \beta, \gamma, \delta)$, so that the relation between the Σ points is of the nature of that above considered. The relation between (x, y, z, w) is given by the equation

$$J(S, T, U, V) = 0;$$

viz., the locus is the Jacobian of the 4 quadric surfaces. The relation between $(\alpha, \beta, \gamma, \delta)$ is given by the equation

$$\nabla = \begin{vmatrix} a\alpha + a_1\beta + a_2\gamma + a_3\delta, & h\alpha + \dots, & g\alpha + \dots, & l\alpha + \dots \\ h\alpha + \dots, & b\alpha + \dots, & f\alpha + \dots, & m\alpha + \dots \\ g\alpha + \dots, & f\alpha + \dots, & c\alpha + \dots, & n\alpha + \dots \\ l\alpha + \dots, & m\alpha + \dots, & n\alpha + \dots, & d\alpha + \dots \end{vmatrix} = 0;$$

so that the locus is (by the foregoing definition) the symmetroid. And the determinant and the symmetroid thus corresponds to the cone-vertex on the Jacobian.

72. But the Jacobian may be obtained in a different manner; viz., if we establish the equations

$$\begin{aligned} &(\xi\partial_a + \eta\partial_b + \zeta\partial_c + \omega\partial_d)S = 0, \\ &(\quad\quad\quad\text{do.}\quad\quad\quad)T = 0, \\ &(\quad\quad\quad\text{do.}\quad\quad\quad)U = 0, \\ &(\quad\quad\quad\text{do.}\quad\quad\quad)V = 0, \end{aligned}$$

then the elimination of $(\xi, \eta, \zeta, \omega)$ leads to the equation $J(S, T, U, V) = 0$ of the Jacobian. And since each of the equations is symmetrical in regard to (x, y, z, w) , C. VII. 21

73. The points (x, y, z, w) of points of $(\alpha, \beta, \gamma, \delta)$, viz., we may either of these latter ones taking the point (x, y, z, w) on the Jacobian surface, the point $(\alpha, \beta, \gamma, \delta)$ on the symmetroid is related to the point (x, y, z, w) , and conversely the system of 4 points, Q on the symmetroid, whose centres of these 4 points respectively; or the equation of the symmetroid, when referred to the 4 points, Q on the symmetroid, P and P' on the Jacobian. Q. V. S. I.

74. Consider any 5 pairs of points (x_1, y_1, z_1, w_1) , $(\xi_1, \eta_1, \zeta_1, \omega_1)$, &c., related as above; the quartic surfaces $S = 0$, $T = 0$, $U = 0$, $V = 0$ are by the general quadric surface which cuts harmonically the five pairs of points respectively; or say they are quadric surfaces *harmonically related* to the 5 pairs of points respectively. The general quadric surface $\alpha S + \beta T + \gamma U + \delta V = 0$, harmonically related to the 5 pairs, will be a cone if $(\alpha, \beta, \gamma, \delta)$ is on the symmetroid, and the vertex of the cone is then defined by M . Chasles (*Comptes Rendus*, tom. LXI., 1865, p. 1147–62), and shown by him to be a variable surface, is thus identified with the Jacobian of any 4 quartic surfaces; and indicated herein we have the particular case, also considered by him, of the locus of the vertices of the quadric cones which pass through 6 given points, or Jacobian of the 6 given points.

75. It is to be shown that there are 10 systems of values $(\alpha, \beta, \gamma, \delta)$, for which the symmetroid $\nabla = 0$ is satisfied; viz., the number of conditions that there should be a node (x, y, z, w) are sufficient: the equation $\alpha S + \beta T + \gamma U + \delta V = 0$ of the symmetroid seems to contain homogeneously 40 constants. But starting with any given symmetrical determinant, the line by a constant determinant, and then column into column by another constant determinant; the coefficients of the constant determinants would in this manner be used, but the resulting product is a symmetrical determinant; and the equation is then with 10 nodes, contains $34 - 10 = 24$ constants.

Symmetroid with given Nodes.

76. A symmetroid can be formed with 7 given points as nodes; but there is no proper symmetroid with 8 given points as nodes. If we endeavour to form such a symmetroid, we obtain a system of 2 quartic cones, each of them passing through the 8 points; the curve of intersection is then a sextic C_2 found in the plane of the conditions of having a given point with the 7 points; the additional condition $(x, y, z, w) = 0$ may be expressed as a linear identification with the conditions of nodality; and also that it is the lineo-linear transformation.

77. Applying the conclusion to the system of quadric surfaces $\alpha S + \beta T + \gamma U + \delta V = 0$, we constants. But starting with this shall be a plane-pair implies a threefold relation between the coefficients A, B , &c., and the required number of nodes is equal to the order of this threefold relation. Establishing the conditions A, B , &c., any 6 linear relations whatever, we should have a ninefold relation between the coefficients of the number of solutions would be equal to the order of the threefold relation; and the number of solutions is clearly = 10.

78. The equation $\nabla = 0$ of the symmetroid seems to contain homogeneously 40 constants. But starting with any given symmetrical determinant, we may multiply it line into line by a constant determinant, and then column into column by another constant determinant, in such wise that the resulting product is still a symmetrical determinant; and the coefficients of the constant determinant may then be used to specialize the form of the equation. The equation $\nabla = 0$ of the symmetroid contains, therefore, the constant determinant; the term $(\alpha, \beta, \gamma, \delta) \Sigma \sigma x_i = w^2 - 1$, $\Sigma x x_i = 3a - 3$, but not every rational determinant; the number of constants in $\nabla = 0$ may then be used to specialize the form of the equation. The equation $\nabla = 0$ of the symmetroid contains, therefore, $40 - 16 = 24$ constants; this is as it should be, for the symmetroid, qua quartic surface with 10 nodes, contains $34 - 10 = 24$ constants.

Symmetroid with given Nodes.

79. A symmetroid can be formed with 7 given points as nodes; but there is no *proper* symmetroid with 8 given points as nodes. If we endeavour to form such a symmetroid with 8 given points as nodes, we obtain a system of 2 quartic cones, each of them passing through the 8 points: the curve of intersection is in fact a quartic curve, C_4 in the plane Π, Π ($r = 4$). It is clear, from this consideration, that any one of the conditions for the nodality of any one of the 8 points is in fact deducible from the others, leaving us with 7 conditions; and a symmetroid with 7 given points as nodes can be formed: but I have not succeeded in effecting this.

80. We have for any node $(\alpha, \beta, \gamma, \delta)$ of the symmetroid,

$$\alpha S + \beta T + \gamma U + \delta V = \text{plane-pair.}$$

If, then, 4 of the given nodes are $(1, 0, 0, 0)$, $(0, 1, 0, 0)$, $(0, 0, 1, 0)$, $(0, 0, 0, 1)$, we must have S , T , U , V each of them a plane-pair. We may without loss of generality assume $S = x^2 + y^2$, $T = x^2 + w^2$; this, however, does not determine the signification of the coordinates (x, y, z, w) ; viz., we have S to determine the meaning of x, y ; and similarly T to determine the meaning of x, w ; and similarly T remain unaltered if we write

$$x \cos \theta + y \sin \theta, \quad x \sin \theta - y \cos \theta \quad \text{for } x, y;$$

and similarly T will remain unaltered if we write

$$x \cos \theta_1 + w \sin \theta_1, \quad x \sin \theta_1 - w \cos \theta_1 \quad \text{for } x, w.$$

Hence, if we go on to assume

$$\begin{aligned} U &= k(x + my + nz + pw)(x + m'z + n'z + p'w), \\ V &= l_1(x + m_1y + n_1z + p_1w)(x + m'_1z + n'_1z + p'_1w), \end{aligned}$$

formulae which contain the 12 constants

$$(k, m, n, p, m', n', p', l_1, m_1, n_1, p_1, m'_1, n'_1, p'_1),$$

this is right, for the symmetroid containing 24 constants, the symmetroid with 4 given nodes contains $(24 - 4 \cdot 3 =) 12$ constants. And each additional condition gives a single one of these constants; and so on.

81. But for any 4 new nodes, the equations may be satisfied by writing therein $u = u', p = -p', m = -m_1, n_1 = n'_1$; viz., they thus assume the form

$$\begin{aligned} S &= x^2 + y^2, \\ T &= x^2 + w^2, \\ U &= (\alpha x + cx)^2 + (by + dw)^2, \\ V &= (\alpha'x + c'x')^2 + (b'y + d'w)^2, \end{aligned}$$

containing 8 constants, which may be determined to make the surfaces have 4 given points. If now with the last mentioned values we form the value of $\alpha S + \beta T + \gamma U + \delta V$, this is a plane-pair if a square if

$$(\alpha + \gamma a^2 + \delta a'^2)(\beta + \gamma c^2 + \delta c'^2) - (\gamma ac + \delta a'c')^2 = 0, \quad \text{say this is } \Lambda = 0,$$

(continued)

82. (continued)

and the second will be a square if

$$(\alpha + \gamma b^2 + \delta b'^2)(\beta + \gamma d^2 + \delta d'^2) - (\gamma bd + \delta b'd')^2 = 0, \quad \text{say this is } \Lambda' = 0;$$

so that the condition

$$\alpha S + \beta T + \gamma U + \delta V = \text{cone}$$

will be satisfied if $\Lambda = 0$, or if $\Lambda' = 0$; that is, the equation of the symmetroid will be $\Lambda\Lambda' = 0$, breaks up into the 2 quadric surfaces $\Lambda = 0, \Lambda' = 0$, each of which is a cone.

82. It is to be further observed that, considering the first mentioned 4 points $(1, 0, 0, 0)$, &c., and any other 4 given points whatever, the equation of the 4 quadric cones through 4 given points 1, 2, 3, 4 and one of the new points, are obtained by considering the values $(\alpha, \beta, \gamma, \delta)$ which satisfy $\Lambda = 0$, viz., they are equal to be of the form

$$(*\$ \beta\gamma, \gamma\alpha, \alpha\beta, \beta\delta, \gamma\delta)^2 = 0;$$

this combined with the first form gives, say through $(1, 0, 0, 0)$, &c., and the 5th point of the symmetroid will be satisfied if $\Lambda' = 0$; the equation of the symmetroid will be one of the 4 cones;

viz., we have to express that one of the 4 cones; viz., we have to express the conditions of the constants b, c in the expressions of the foregoing equations; that is the conditions for 4 given points different therefrom by changing a, b, c into $a + \frac{1}{w}(by - gz)$, $b + \frac{1}{w}(-kx + fz)$, $c + \frac{1}{w}(gx - fy)$ respectively; the equation may therefore be written in the symbolic form

$$w^2 \cdot \exp \frac{1}{w} [(ky - gz) \partial_a + (-kx + fz) \partial_b + x^2(gx - fy) \partial_c] \cdot [(a, b, c)^2] = 0,$$

or, what is the same thing,

$$w^2 \cdot \exp \frac{1}{w} [x(g\partial_b - h\partial_c) + y(h\partial_a - f\partial_c) + z(f\partial_b - g\partial_a)] \cdot [(a, b, c)^2] = 0,$$

where $\exp \theta$ (read exponential) denotes e^θ , and $[(a, b, c)^2]$ represents a determinant as above explained. The equation contains, it is clear, the four terms

$$x^2[(a, -h, g)^2] + y^2[(-h, b, -f)^2] + z^2[(g, -f, c)^2] + w^2[(a, b, c)^2].$$

I am not sure whether this surface of the eighth order has been anywhere considered.

83. In order that the symmetroid may have a 6th given node $(\alpha_1, \beta_1, \gamma_1, \delta_1)$, I observe that the constants may be determined so that $\alpha_1 S + \beta_1 T + \gamma_1 U + \delta_1 V$ shall be equal to the variable factor, writing this as Π^2 and the like for the 5th point; we find $a(\alpha_1 + 5\alpha_1 + 6\alpha_1) + \dots = 0$, viz., the new node being $(\alpha_1, \beta_1, \gamma_1, \delta_1)$, and putting therein $a_i(1 + i\alpha_i + 6\alpha_1) = 0$, &c., for the various values of values which the new nodes can assume the formulae

$$\alpha_i S + \beta_i T + \gamma_i U + \delta_i V = (a_i, b_i, c_i \times x, y, z, w)^2,$$

the new variables given through 4 given nodes $(1, 0, 0, 0)$, &c., and the points $1, 2, \dots, 8$ points; the given plane-pairs at the points $1, 2, \dots, 7$.

Quartics with 11 or more Nodes.

103. I mention two results which, although they relate to quadric surfaces with more than 10 nodes, present themselves in such immediate connexion with the present Memoir, that it is natural to speak of them. If, in the equation

$$\begin{vmatrix} A & H & G & L \\ H & B & F & M \\ G & F & C & N \\ L & M & N & D \end{vmatrix} = 0$$

of the symmetroid $(A, B, \dots$ linear functions of the coordinates), we have identically $A = 0$, then the surface has evidently a node in the ordinary manner in addition to the usual 10 nodes; an the surface has in all 11 nodes. And so also if (identically in every case) $B = 0$, there are 12 nodes; if $C = 0$, there are 13 nodes. These are, in fact, quartic surfaces with 11, 12, 13, and 14 nodes respectively, mentioned in Kummer's Memoir.

104. We may consider the symmetroid derived from the quadric surfaces which pass through 6 given points: viz., taking as before (see No. 25) the conditions of the 6 points to be $(1, 0, 0, 0)$, $(0, 1, 0, 0)$, $(0, 0, 1, 0)$, $(0, 0, 0, 1)$, $(1, 1, 1, 1)$, (a, b, c, d) ; the coordinates of the line joining the last-mentioned two points: and, to avoid confusion, taking for the present purpose (X, Y, Z, W) instead of $(\alpha, \beta, \gamma, \delta)$ for the coordinates of a point on the symmetroid, the equation is obtained by arranging in the form of a determinant the coefficients of the quadric form

$$\begin{aligned} & Xx(\quad \quad \quad) + hy(X - Y) + cyW, \\ & h(X - Y) + cyW, \\ & g(Z - X) + byW, \quad f(Y - Z) + axW, \\ & W(axyx + b\beta zx + cyxy \quad \quad \quad); \end{aligned}$$

viz., the equation in question is

$$\begin{vmatrix} \cdot & h(X - Y) + cyW & g(Z - X) + byW & aX \\ h(X - Y) + cyW & \cdot & f(Y - Z) + axW & bY \\ g(Z - X) + byW & f(Y - Z) + axW & \cdot & cZ \\ aX & bY & cZ & \cdot \end{vmatrix} = 0,$$

or, as it may be more simply written,

$$\sqrt{aX[f(Y - Z) + axW]} + \sqrt{bY[g(Z - X) + byW]} + \sqrt{cZ[h(X - Y) + cyW]} = 0.$$

This is, in fact, a surface with 16 nodes. It would appear that additional nodes correspond to the six common nodes that the 4 quadric surfaces having in common 1, 2, 3, 4, 5, or 6 points, the corresponding symmetroid would have 11, 12, 13, 14, 15, or 16 nodes. But I reserve this for future consideration.

105. To find the equation to the quadric surface through the three lines $(\alpha_1, \beta_1, \gamma_1, f_1, g_1, h_1)$, $(\alpha_2, \beta_2, \gamma_2, f_2, g_2, h_2)$, $(\alpha_3, \beta_3, \gamma_3, f_3, g_3, h_3)$, $(\alpha_4, \beta_4, \gamma_4, f_4, g_4, h_4)$, $(\alpha, \beta, \gamma, f, g, h)$. Take the equation of a quadric surface through this line will be of the form

$$\begin{vmatrix} x^2 & y^2 & z^2 & w^2 & yz & zx & xy & yw & zw & xw \\ \alpha^2 & \beta^2 & \gamma^2 & f^2 & \gamma\beta & \gamma\alpha & \alpha\beta & h\gamma & f\beta + g\alpha & f\gamma \\ \alpha_1^2 & \beta_1^2 & \gamma_1^2 & f_1^2 & \gamma_1\beta_1 & \gamma_1\alpha_1 & \alpha_1\beta_1 & h_1\gamma_1 & h_1\alpha_1 & h_1\beta_1 \\ 2\alpha_1\alpha & 2\beta_1\beta & 2\gamma_1\gamma & 2f_1f & f\gamma_1 + f_1\gamma & f\alpha_1 + f_1\alpha & \frac{1}{2}h(g_1 + gh_1) & f_1h + fh_1 & f_1g + fg_1 & \frac{1}{2}h(g + \beta h_1) \\ \alpha_2^2 & \beta_2^2 & \gamma_2^2 & f_2^2 & \gamma_2\beta_2 & \gamma_2\alpha_2 & \alpha_2\beta_2 & h_2\gamma_2 & h_2\alpha_2 & h_2\beta_2 \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \end{vmatrix} = 0$$

and we form thus a determinant with three of its lines relating to the line 1, three of them to the line 2, and three to the line 3, and the equation of the quadric surface through the lines 1, 2, 3; and it is clear that the determinant of the order $9 \cdot 9$ contains, as a factor, the principal counter-system of the second plane, and the corresponding curves the principal counter-system of the first plane.

106. Taking $(\xi, \eta, \zeta, \omega)$ as plane-coordinates, two quadric surfaces

$$\begin{aligned} & (a, b, c, d, f, g, h, l, m, n) \chi (\xi, \eta, \zeta, \omega)^2 = 0, \\ & (a_1, b_1, c_1, d_1, f_1, g_1, h_1, l_1, m_1, n_1) \chi (\xi, \eta, \zeta, \omega)^2 = 0 \end{aligned}$$

are said to be intervorts (or interverse) one of the other, when we have between the coefficients the relation

$$(a, b, c, d, f, g, h, l, m, n) \times (a_1, B_1, C_1, F_1, G_1, H_1, L_1, M_1, N_1) = 0,$$

that is,

$$aA + \dots + 2fF + \dots = 0.$$

The condition that the two surfaces may be intervorts of each other is linear in regard to the coefficients of each of them; hence, taking a before explained location, intervorts to two given quadrics A quadric, we have 9 quadrics; intervorts to 5 given quadrics is a given quadric surface, we have between the coefficients $10 - k$ quadrics, that is, $5 - k$ moreover, and the question is easily supplied.

If the quadric $(a, \dots) \times (\xi, \eta, \zeta, \omega)^2 = 0$ be a plane-pair, the plane-coordinates of the planes are by the equations

$$\begin{aligned} (\partial_a + ay + az + pw) \partial_\xi + p_1 \partial_\eta &= 0, \\ hx + \dots, \quad bx + \dots, \quad fx + \dots, \quad mx + \dots &= 0, \\ gx + \dots, \quad fx + \dots, \quad nx + \dots &= 0, \\ lx + \dots, \quad mx + \dots, \quad nx + \dots, \quad dx + \dots &= 0; \end{aligned}$$

viz., the locus is (by the foregoing definition) the symmetroid. And the determinant and the symmetroid corresponds to the cone-vertex on the Jacobian.

107. Taking the lines to be those of $(a_1, b_1, c_1, f_1, g_1, h_1), (a_2, b_2, c_2, f_2, g_2, h_2) \dots (a_5, b_5, c_5, f_5, g_5, h_5)$ to be those of three quadric surfaces having no common conditions: But if the conditions $(\xi, \eta, \zeta, \omega)$ belonging to a point, the value is $= 0$.

108. Any 6 points whatever may be regarded as points on a skew cubic; and the coordinates (x, y, z, w) may then be taken to be of the form

$$\begin{vmatrix} x & y & z \\ y & z & w \end{vmatrix} = 0. \quad \text{This being so, the coordinates of the 6 given points may be taken to be } (1, t_1, t_1^2, t_1^3), (1, t_2, t_2^2, t_2^3),$$

the 6 points; that is, of the surface containing the conditions $(1, t_3, t_3^2, t_3^3), (1, t_4, t_4^2, t_4^3)$; and the equation of the Jacobian surface of

109. Representing as before each line by means of its six coordinates, let $(\alpha, \beta, \gamma, f, g, h)$ be the coordinates of the vertex, the coordinates (X, Y, Z, W) be the coordinates of any one of the lines, the equation (a, b, c, f, g, h) the coordinates of any one of the six lines, the equation of the plane through this line and the vertex for shortness

$$\alpha(zW - wY) + b(yW - wY) + c(xW - wX) = 0;$$

or, what is the same thing, putting for shortness

$$\begin{aligned} P &= \cdot - hy - gz + aw, \\ Q &= -hx \cdot + fz + bw, \\ R &= gx - fy \cdot + cw, \\ S &= -ax - by - cz \cdot, \end{aligned}$$

the equation is

$$PX + QY + RZ + SW = 0.$$

The plane in question is a tangent plane to the cone touched by each of the 6 lines. Now when the plane is a tangent plane to the cone touched by each of the 6 lines, the plane in question is a tangent plane to the cone touched by each of the 6 lines. Hence taking the plane $PX + QY + RZ + SW = 0$ on the cone touched by the 6 lines, the equation

$$PX + QY + RZ + SW = 0.$$

To ascertain the form of this, write for a moment $y = 0, x = 0$; the equation is

$$[(aw + bz + bw, \quad yz + ew)^2] = 0,$$

or attending only to the highest and lowest powers of w , this is

$$w^2[(\alpha, b, c)^2] + \dots + w^4[(a, -h, g)^2] \dots$$

and it is thence easy to infer that the whole equation divides by w^2 , so that, omitting this factor, the form of the equation is

$$[(a, b, c \times g, y, z, w)^2 \dots] = 0;$$

viz., this equation is satisfied if z denote any one of the quantities $(\alpha_1, \beta_1, \gamma_1, f_1, g_1, h_1)$, &c., that is, if any one of the 6 lines pass through the assumed point in the direction of any vertex of the cone, and if so, the equation of the cone, regarded as a function of w , omitting this factor, the form of the equation is

$$((a, b, c, f, g, h \times x, y, z, w)^2 = 0,$$

viz., the equation is satisfied if z denote any of the quantities $\alpha_1, b_1, c_1, f_1, g_1, h_1$; and if so, the equation of the cone, regarded as a function of x, y, z, w , is

$$\begin{vmatrix} x, & y, & z \\ a, & b, & c \end{vmatrix} = 0.$$

Locus of the vertex of a Quadric Cone which touches each of Six given Lines.

110. Representing as before each line by means of its six coordinates, let $(\alpha, \beta, \gamma, f, g, h)$ be the coordinates of the vertex, and (X, Y, Z, W) the current coordinates of any one of the lines, then the equation of the plane through this line and the vertex is

$$\alpha(zW - wZ) + b(yW - wY) + c(xW - wX) + f(yZ - zY) + g(zX - xZ) + h(xY - yX) = 0;$$

or, in the same notation as before, for shortness

$$\begin{aligned} P &= \cdot - hy - gz + aw, \\ Q &= -hx \cdot + fz + bw, \\ R &= gx - fy \cdot + cw, \\ S &= -ax - by - cz \cdot, \end{aligned}$$

the equation is

$$PX + QY + RZ + SW = 0.$$

The plane in question is a tangent plane to the cone touched by each of the 6 lines. Hence taking the plane $PX + QY + RZ + SW = 0$ on the cone touched by the 6 lines, attending only to the highest and lowest powers of w , this is

$$w^2[(a, b, c)^2] + \dots + w^4[(a, -h, g)^2] \dots,$$

and it is thence easy to infer that the whole equation divides by w^2 , so that, omitting this factor, the form of the equation is

$$((a, b, c, f, g, h) \times (x, y, z, w))^2 = 0,$$

viz., the equation in question is of the order 8 in the coordinates (x, y, z, w) and of the degree 2 in the coordinates of each of the lines. It would not be very

111. difficult to actually develop the equation; in fact, starting from the term $w^2[(a, b, c)^2]$ the other terms are obtained therefrom by changing a, b, c into $a + \frac{1}{w}(hy - gz)$, $b + \frac{1}{w}(-hx + fz)$, $c + \frac{1}{w}(gx - fy)$ respectively; the equation may therefore be written in the symbolic form

$$w^2 \cdot \exp \frac{1}{w}[(hy - gz) \partial_a + (-hx + fz) \partial_b + (gx - fy) \partial_c] \cdot [(a, b, c)^2] = 0,$$

or, what is the same thing,

$$w^2 \cdot \exp \frac{1}{w}[x(g \partial_b - h \partial_c) + y(h \partial_a - f \partial_c) + z(f \partial_b - g \partial_a)] \cdot [(a, b, c)^2] = 0,$$

where $\exp \theta$ (read exponential) denotes e^θ , and $[(a, b, c)^2]$ represents a determinant as above explained.

112. I am not sure whether this surface of the eighth order has been anywhere considered.

[446] ON THE MECHANICAL DESCRIPTION OF A NODAL BICIRCULAR QUARTIC.

[From the *Proceedings of the London Mathematical Society*, vol. III. (1869–1871), pp. 100–106.]

The ingenious method, devised by Mr S. Roberts (*Proceedings*, vol. II. p. 133) for the description of a nodal bicircular quartic suggests a further investigation. We have a quadrilateral $OAA'O'$, in which the adjacent sides OA, AA' are equal to each other, and the other two adjacent sides $OO', O'A'$ are also equal to each other; O, O' are fixed points; and we have thus a link AA' , the extremities of which are connected by means of the radii $OA, O'A'$ respectively, and consequently describe circles about the centres O, O' respectively, the radius OA of the one circle being equal to the length AA' of the link, and the radius $O'A'$ of the other circle being equal to the distance OO' of the centres. The theorem is, that any point C , rigidly connected with the link AA' , describes a nodal bicircular quartic, that is, a quartic curve with three nodes (or unicursal quartic), two of the nodes being the circular points at infinity. Any such curve is the inverse of a conic, and it is also the antipode of a conic; viz., if at such point of the curve we draw a line at right angles to the radius vector from the node, then envelope of the line passing through this point is a circle. If the equalities $OA = AA', O'A' = OO'$ did not subsist, then to a given position of the link AA' corresponds two positions of A, A' respectively, on the circles; the radius OA of the one circle being equal to the length AA' of the link, and the radius $O'A'$ of the other circle being equal to the distance OO' . The theorem is, that any point C , rigidly connected with the link AA' , describes a nodal bicircular quartic, that is, a quartic curve with three nodes (or unicursal quartic), two of the nodes being the circular points at infinity.

I have a curve AA' : the link; OO' may be called the bar; and the locus of the radius AA' produced; moreover AA' may be called the link; the radius OA of the one circle being equal to

the length AA' , and the radius $O'A'$ of the other circle being equal to the distance OO' of the centres of the two circles respectively. Suppose the link AA' to slide along the bar OO' ; the link AA' remains parallel to itself in this its motion; viz., on this slide-motion the bar OO' moves about its centre, and the link AA' traverses the constant triangle; and, producing OA , $O'A'$ to meet in K , the locus of K is called the constant triangle; and the locus of C will be a higher order than is in the actual problem.

I have called AA' the link; OO' may be called the bar. OA is then the link-radius, $O'A'$ the bar-radius; moreover AA' may be called the constant triangle; and, producing OA , $O'A'$ to meet in K , then AKA' is the constant triangle. Since at any instant the motion of A is normal to KA , and the motion of A' to KA' , it is clear that the motion of any point on KA or on KA' is also normal to KA , KA' respectively; and in like manner the motion of A' , in its position on the line KA' , is in like manner the motion of any point on the line KA' . We have thus an instantaneous rotation about the point K .

Imagine any two positions of the link; say these are $A_1A'_1$, and $A_2A'_2$. Join A_1A_2 and at its mid-point draw a perpendicular thereto; join in like manner $A'_1A'_2$, and at its mid-point draw a perpendicular thereto; and let these two perpendiculars meet in T ; we have the two equal triangles $TA_1 = TA_2$, $TA'_1 = TA'_2$, and the angles $A_1TA'_1 = A_2TA'_2$, so that there is a rotation about the centre T , which conducts $A_1A'_1$ into $A_2A'_2$. Now, in the limit, when the two positions $A_1A'_1$, $A_2A'_2$ are made to coincide with the triangles $A_1A'_1T$ and $A_2A'_2T$ respectively; that is, T will be a node in the locus described by this point, but in the present case, this point K is a fixed point. It is indifferent any point will pass with the link AA' in some such manner, and not indifferent any point of the locus C , of which the same considerations may be applied to itself; it is, for any point of the locus, when the link describes the ordinary points of curvature. And in the linkage describing the bicircular quartic, the centres are at the same distance, a from the centre O of the linkage. The quartic, when described by the two points A_1A_2 describes a circle of the same diameter centred at O of the linkage and centres at the same distance from the centre O of the linkage; the quartic curve, when described by the two points A_1A_2 describes a circle of the same diameter a from the centre O of the linkage and centres at the same distance from O .

Passing now to the analytical investigation, I take the origin at O , the axis of x along the direction from O to O' , and the axis of y , at right angles thereto, upwards from O . The inclinations of OA , AA' , $O'A'$ to the axis Ox are taken to be θ , ϕ , θ' respectively. I write also $OA = AA' = a$, $OO' = O'A' = a'$; and the angles θ , θ' are connected by the relation $\sin \theta' / \sin \theta = a/a'$, and finally $AB = b$, $BC = c$.

Observing that the angle $AA'O'$ is $= \theta$, we have $\theta' = \pi - \theta - \phi$, or $\theta - \phi$ is the deviation from O , and the axis of y , at right angles, upwards from O . The inclinations of OA , AA' , $O'A'$ to the axis Ox are taken to be θ , ϕ , θ' respectively. I write also $OA = AA' = a$, $OO' = O'A' = a'$; and finally $AB = b$, $BC = c$.

Hence, if we go on to assume

$$m = 1 : 1 + m : 1 - m = a' : a : a' + a : 2a' : 2a;$$

and finally $AB = b$, $BC = c$.

Observing that the angle $AA'O'$ is $= \theta$, we have $\theta' = \pi - \theta - \phi$, upwards from O . The inclinations of OA , AA' , $O'A'$ to the axis Ox are θ , ϕ , θ' respectively. I write also $OA = AA' = a$, $OO' = O'A' = a'$; and $OO' = a$, $OO' = a'$; and finally $AB = b$, $BC = c$.

(Algebraic developments.) For the locus of C we have

$$x = a \cos \theta + b \cos \phi + c \sin \phi,$$

$$y = a \sin \theta + b \sin \phi + c \cos \phi,$$

or, instead of θ, ϕ introducing u , we have

$$x = -a \frac{u^2 - 1}{u^2 + 1} + b \frac{u^2 - m^2}{u^2 + m^2} - c \frac{2mu}{u^2 + m^2},$$

$$y = a \frac{2u}{u^2 + 1} + b \frac{2mu}{u^2 + m^2} + c \frac{u^2 - m^2}{u^2 + m^2}.$$

[Further reductions and the curve obtained as an inverse-of-a-conic and antipode-of-a-conic are given in the remainder of the paper.]

(Closing remarks.) There remains to be observed, further, that the equation of the line in question obtained by writing in the foregoing equations, $u = u'$ and eliminating the remaining quantities δx , δy , &c., and the corresponding expressions δx_i , δy_i , &c.; the result of the elimination will be the same as if, writing $u = u_1$, we have in fact only to consider the equation of the line in the form

$$ax' + by' + c = 0 \quad (\text{in fact, for } A, B \text{ or of } (x, y).)$$

The remainder of the paper carries out this elimination and shows that the locus is in fact a circular quartic with the prescribed nodal properties.

[447] ON THE RATIONAL TRANSFORMATION BETWEEN TWO SPACES.

[From the *Proceedings of the London Mathematical Society*, vol. III. (1869–1871), pp. 127–180.
Account of the Paper given at the Meeting 11 March 1869.]

Two figures are rationally transformable each into the other (or, say, there is a rational transformation between the two figures) when to a variable point of each of them there corresponds a single variable point of the other. The figures may be either lost in a space, or loci of any number of dimensions; or they may be lost in plane spaces themselves. Thus the figures may be each a line of one space (or space of one dimension); and there will be a rational transformation between two lines, or they may be each a plane (or space of two dimensions); and there will be a rational transformation between two planes; or, finally, the figures may be each surfaces (or spaces of three dimensions), and there will be a rational transformation between two spaces. The problem of the rational transformation between two planes (or generally between two plane spaces of equal dimensions) is the only one which has been hitherto considered, and it is in fact thereto, that every rational transformation between two planes, and to be more particular still, between two plane spaces, may be reduced. The general problem in three dimensions is here treated of; the most general transformation between two plane spaces of equal dimensions is, in fact, the transformation (which is already mentioned) in the only one peculiar transformation between two plane spaces; and consequently a rational transformation between two planes; viz., in regard hereto I examine the general theory, but attend mainly to what I call the lineo-linear transformation; viz., it is assumed that the coordinates

The General Principle of the Rational Transformation between Two Spaces.

1. In all that follows, the two spaces (lines, planes, or three dimensional spaces, as the case may be), to any rational corresponding loci in the two spaces respectively, are referred to as the first and second figures respectively; viz., between two lines, between two planes, or between two three-dimensional spaces, as the case may be. The notation of the first and second figures and the corresponding loci, when considered, not as superimposed or situate in a common space, but as existing, each independently of the other, but as together mentioned in the only one peculiar way, will be referred to as the first and the second figures and the corresponding loci. Writing as before

$$x : y : z = X : Y : Z,$$

the unicursal correspondence (which is, in fact, the only one between two lines) unicursal correspondence; viz., if as before, we refer to the lines, planes, or three-dimensional spaces, the correspondence between them is essentially unsymmetrical.

2. If, to fix the ideas, we attend to the case of two planes, or take the sets

$$x : y : z = X : Y : Z, \quad x : y : z = X' : Y' : Z',$$

[The opening articles, §§ 2–35 on PDF pages 212–225, develop the general theory of the rational transformation between two spaces, in particular establishing the basic formulæ of the lineo-linear (or, as Cayley calls it, the homographic) transformation between two planes; the construction of the constants of the transformation; the conditions under which a system of equations of the rational correspondence is satisfied; the count of free constants; the various special and degenerate cases (when the transformation breaks up into a product of a linear and a rational correspondence, or when certain equations of correspondence become identically satisfied); the use of these formulæ for the calculation of the number of intersections of a curve in the first plane with its transform in the second plane; the count of free parameters in the transformation; and, finally, the discussion of the system of *principal curves* and *principal counter-curves* (or *réseaux* of Cremona) which arise in the transformation. The detailed algebraic developments occupy pages 189–203 of the book, and the memoir is continued from book page 204 onward in the next chunk of the typesetting.]

(End of chunk: PDF pages 176–225, book pages 154–203.

Paper 445 §§ 56–112 (tail of the Memoir on Quartic Surfaces),

paper 446 in full,

paper 447 §§ 1–35 (opening of the Rational Transformation memoir).)